Explainability in Reinforcement Learning

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Chapter 1

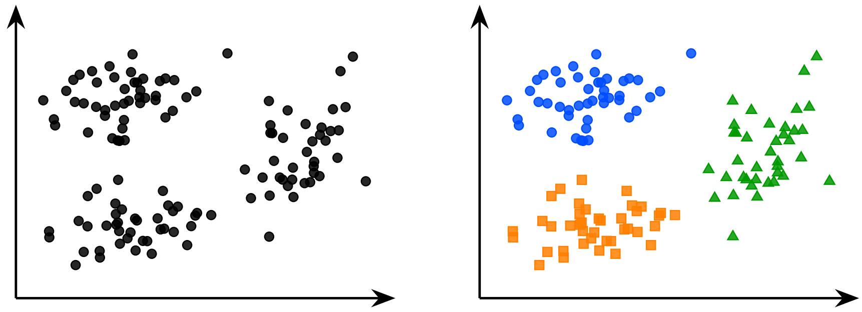
Latent Space Analysis

The ability of a Neural Network (NN) to generalize provides evidence that the model has learned more than a simple memorization of the training examples. Instead, it must construct internal representations that capture regularities present in the data. These representations play a central role in the network's ability to solve a task. Understanding their structure is therefore a key step toward understanding how NN achieve generalization.

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep Learning (DL) introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, NN gradually construct representations that become increasingly relevant to the task under consideration [Rumelhart et al., 1986]. These representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is precisely through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training. Since these representations are learned internally by the network and are not directly observable from the input or output spaces, they are commonly referred to as Latent Representation (LR). The collection of these LR learned throughout the network is commonly referred to as a Latent Space (LS).

These LS are now at the core of a growing number of practical applications. Neural video compression exploits the compactness of LS to encode information efficiently [Lu et al., 2019]. In Reinforcement Learning (RL), agent internal representations are used for efficient exploration in extremely open environments [Burda et al., 2018]. Recommendation systems model users and content within a shared LS to compute meaningful similarities [He et al., 2017]. Retrieval-augmented generation systems index and query knowledge bases through latent embeddings [Lewis et al., 2021]. Finally, Large Language Model (LLM) and multimodal architectures rely on the quality of learned representations to align heterogeneous modalities such as text, images, and audio [Radford et al., 2021]. These examples represent only some of the most widespread applications of LR.

Given their extensive use, the study of LR has become a central topic in DL research. A major objective is to understand how information is organized within these spaces, which concepts are encoded, and how these representations



(a) Observed activations **(b)** Under the concept hypothesis

Figure 1.1: The same latent space activations seen without concept hypothesis **(a)** and under the working hypothesis that concepts emerge in the latent space **(b)**. Colors represent hypothetical concept memberships assigned by the model; they are not observed labels. Under this hypothesis the structure of the latent space becomes meaningful and amenable to analysis.

influence model behavior.

In this chapter, we first make explicit the working hypothesis that underlies much of the research on LS analysis. Throughout this manuscript, we refer to this assumption as the concept hypothesis 1.1. Once this conceptual framework has been established, we review the main approaches proposed in the literature to identify and extract concepts from LR. To the best of our knowledge, the main families of methods include probing methods (Section 1.2), semantic directions (Section 1.3), sparse autoencoders (Section 1.4), and clustering-based approaches (Section 1.5). Finally, Section 1.6 draws a comparative synthesis of these methods and discusses the open questions that remain.

1.1 Concept Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on LS analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, LS, concept, and semantics. The goal of this section is to make this conceptual framework explicit by defining its main components.

This hypothesis originates from the need to explain the generalization ability of deep NN observed after training, as discussed in Section **??**. A model is said to generalize when, for an input $x \in \mathcal{X}$ not seen during training, it still produces an output $\hat{y}$ close to the true output y. Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 1.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task [Bengio et al., 2014].

Definition 1.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction or decision-making.

The ability of NN to perform abstraction arises from their intermediate activations. Indeed, a NN does not directly map x to y. Instead, as illustrated in Figure **??**, it computes a sequence of intermediate activations across hidden layers. These activations correspond to internal encodings of the input, which we refer to as representations in the sense of Definition 1.1.2.

Definition 1.1.2: Latent Representation

A representation is the internal encoding of an input given by a single intermediate layer of a trained NN. While embeddings may exist independently of learning, a representation is a learned embedding shaped by the training process and the task.

Each hidden layer refines the representation of the previous layer. Each transformation progressively reshapes the representation into a form more directly aligned with the output space [Rumelhart et al., 1986]. This process can be described as a sequence of transformations where each $h^{(l)}$ is a representation:

$$x \to h^{(1)} \to h^{(2)} \to \cdots \to h^{(L-1)} \to \hat{y}$$

The set of all such representations produced by a layer over a dataset defines its LS, as defined in Definition 1.1.3.

Definition 1.1.3: Latent Space

The LS is the set of all representations produced by a layer over a dataset. It is shaped by learning and organizes representations according to task-relevant properties.

Under the concept hypothesis, the analysis of the LS is assumed to provide access to higher-level abstract structures learned by the model, as illustrated in **??**. These structures are referred to as concepts. In this manuscript, the term is used in a pragmatic sense. Concepts are not assumed to correspond to objective or immutable entities. Instead, they are viewed as abstract representations that organize information and support reasoning. Their relevance is therefore not determined by whether they constitute the ”true” representation of the system, but by whether they provide a useful and coherent description for the task under consideration [Newell, 1982]. Our definition is given in Definition 1.1.4.

Definition 1.1.4: Concept

A concept is a task-dependent abstraction that captures regularities in the input space relevant for predicting the target output.

This perspective is particularly well aligned with LS analysis, where the objective is to uncover organizing principles that support the model's behavior.

From this perspective, a given representation can be interpreted as a composition of multiple such concepts, each contributing to the final prediction.

The concept hypothesis, as stated above, only assumes that such task-relevant regularities exist within LS and that they can be interpreted as concepts. It does not, by itself, specify how a concept is realized within LS, i.e. under which geometric or algebraic form it should be sought. We refer to this additional assumption as the concept structure, defined in Definition 1.1.5.

Definition 1.1.5: Concept Structure

A concept structure is the specific form assumed for how a concept is instantiated within the LS, for example a linear direction, a subspace, a cluster of points, or a sparse combination of basis features. It determines both the organization attributed to a concept and, consequently, the procedure required to extract it from LR.

Distinguishing the concept hypothesis from the concept structure attributed to it matters because, as detailed below, the methods reviewed in this chapter each commit to a different concept structure.

Another property that can be considered when analyzing concepts is their relationship to human interpretation. This relationship is captured by the notion of semantics [Marconato et al., 2023], defined in Definition 1.1.6. Under this definition, concepts may exhibit varying degrees of semantic content. Some concepts can be readily associated with human-interpretable notions, whereas others may contribute to task performance while remaining difficult to interpret.

Definition 1.1.6: Semantics

Semantics refers to the alignment between concepts in the model LS and concepts used by a human observer. A concept is semantic if it can be consistently associated with a human-interpretable concept.

While the methods reviewed in this chapter all subscribe to the concept hypothesis, they do not agree on the concept structure a concept should take. Probing methods (Section 1.2) assume concepts to be linearly separable directions in LS, semantic directions (Section 1.3) assume concepts correspond to single directions along which representations can be shifted, sparse autoencoders (Section 1.4) assume concepts arise as sparse combinations of learned basis features, and clustering approaches (Section 1.5) assume concepts form discrete regions of LS. As a result, although these methods agree that concepts are present as identifiable structures within LS, they disagree on the exact form these structures take, and consequently on the procedure required to extract them. This divergence is one of the reasons why several, rather than a single, families of methods coexist in the literature, as reviewed in the remainder of this chapter.

It is important to note that this conceptual framework is not supported by a complete theoretical understanding of deep NN. Current mathematical tools do not provide a rigorous characterization of their internal representations and do not formally prove that concepts emerge as identifiable structures within LS, nor that any particular concept structure is the correct one. Nevertheless, this perspective constitutes a widely adopted working hypothesis in the field of

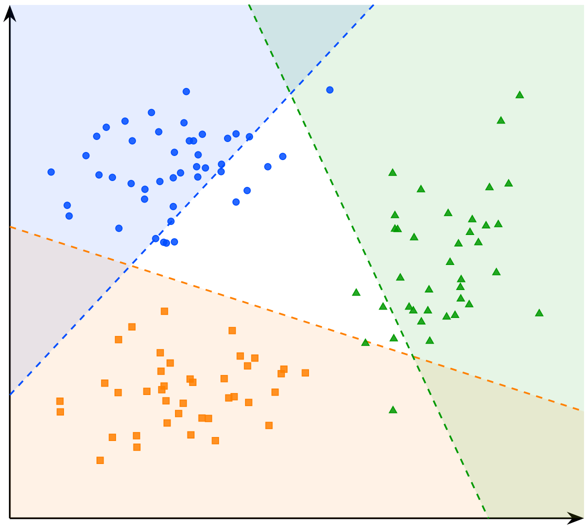


Figure 1.2: Decision boundaries of the linear probes for concepts A, B, and C. The colored regions indicate the areas classified positively by each probe.

representation and LS analysis.

The remainder of this chapter is devoted to reviewing methods for analyzing LS and extracting the concepts they encode. We begin with probing, one of the most direct approaches for assessing whether concepts are represented in the LS.

1.2 Probing

Probing methods aim to determine whether a given semantic concept is encoded within the LS of a NN [Li et al., 2024, Karvonen, 2024]. More specifically, they investigate whether the presence or absence of a selected concept can be inferred from embeddings. The underlying assumption is that if a concept can be reliably decoded from these representations, then the network likely encodes information related to it in order to perform its task.

A probing analysis begins by identifying a set of semantic concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concepts. We denote by c_j the presence or absence of concept j in example x. The dataset can therefore be written as follows:

$$\mathcal{D} = \{(x^{(i)}, y^{(i)}, c_1^{(i)}, \ldots, c_n^{(i)})\}_{i=1}^{N}.$$

The data are subsequently projected into the LS of the pretrained model. We denote by h^l the representation vector obtained at layer l. From these LR, a classifier is trained in a supervised manner to predict the presence of the considered concept. This classifier is referred to as a probe and is denoted by p. Its purpose is to determine whether a given concept is present in the representation. Its purpose is therefore to assess whether information associated

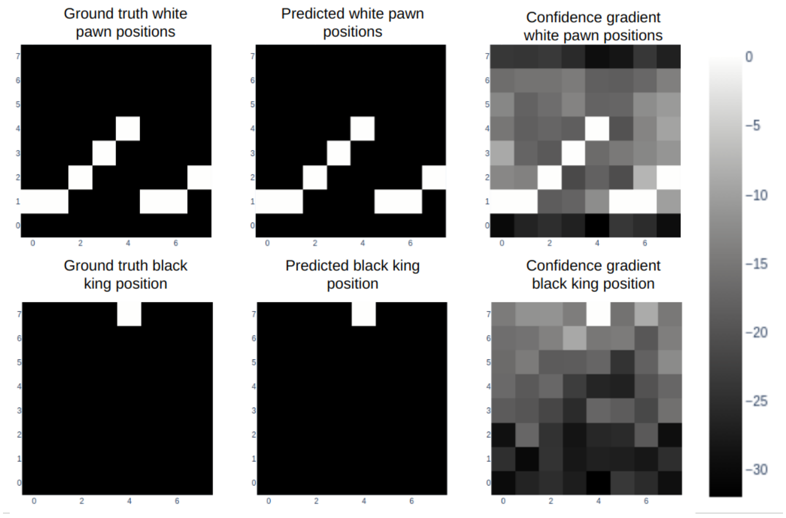


Figure 1.3: Probing analysis of a large language model trained to play chess [Karvonen, 2024]. Each row corresponds to a piece type: white pawns (top) and the black king (bottom). The left column shows the ground truth board positions, the middle column shows the positions predicted by the probe from the model's internal representations, and the right column shows the confidence gradient across board squares. The close agreement between ground truth and predicted positions suggests that the model internally maintains a structured representation of the board state, even though it was only trained on sequences of moves expressed in natural language.

with the concept is accessible from the network's internal representations. This can be formalized as follows:

$$p_\theta^{l,j} : h^l \mapsto c_j, \quad \theta^* = \arg\min_\theta \mathcal{L}\big(p_\theta^{l,j}(h^l), c_j\big)$$

The classifier used to detect the presence or absence of a concept must remain intentionally simple. Indeed, if the probe has too much capacity, it may reconstruct the target information from complex correlations that do not necessarily correspond to information explicitly encoded in the LS. In such a case, it becomes difficult to determine whether the concept is genuinely represented by the network or merely reconstructed through the expressive power of the classifier. For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the NN and potentially usable during decision-making. If the classifier achieves performance significantly above chance level, this suggests that the LS contains exploitable information related to the considered concept.

An application of this technique can be found in studies investigating whether LLM develop internal representations of their environment [Li et al., 2024, Karvonen, 2024]. In these works, researchers train LLM to play games such as chess or Othello in a purely textual setting. The model is provided only with sequences of moves and information about the actions it can take, all expressed in textual form. The central question is whether the LLM develops an internal representation of the game state. To address this question, the probed concepts correspond to the presence or absence of different game pieces on specific board positions. As illustrated in Figure 1.3, probes are able to reconstruct the full board state from the model's hidden representations, even though the only available information to the model is a sequence of text.

However, the fact that a concept can be decoded from LR does not necessarily imply that the model actually uses this information to perform its task. To assess whether representations play a causal role in the model's output, it is possible to perform interventions on the LR. An intervention consists of modifying a LR in a controlled manner by activating or deactivating a given concept, and then studying its impact on the model's output. If the resulting change in the model's output is consistent with the applied perturbation, this suggests that the representation had a causal role.

To this end, the LR h^l is minimally modified such that the probe prediction for the target concept c_j changes state, while the predictions associated with other concepts remain unchanged. This goal can be formulated as an optimization problem over the latent activation itself. The optimization is performed directly on the representation $h^{l'}$, using gradient-based optimization as introduced in Section **??**. The solution to this problem is the optimal modified representation h^{l*}, defined as:

$$h^{l*} = \arg\min_{h^{l'}} \underbrace{\gamma\, \mathcal{L}_{\text{flip}}\big(p_\theta^{l,j}(h^{l'}), c_j\big)}_{\text{flip target concept } c_j} + \underbrace{\lambda \sum_{k\neq j} \mathcal{L}_{\text{preserve}}\big(p_\theta^{l,k}(h^{l'}), c_k\big)}_{\text{preserve other concepts } c_{k\neq j}} + \underbrace{\beta \|h^{l'} - h^l\|_2^2}_{\text{minimality}}$$

where:

- h^l is the original LR of the input at layer l,
- $h^{l'}$ is the modified LR being optimized,
- h^{l*} is the optimal modified representation,
- $\mathcal{L}_{\text{flip}}$: loss encouraging the prediction of the target concept c_j to change state (i.e., flipping the predicted label),
- $\mathcal{L}_{\text{preserve}}$: loss enforcing invariance of all non-target concepts c_k for $k \neq j$,
- $\|h^{l'} - h^l\|_2^2$: regularization term ensuring the modified representation remains close to the original LR,
- γ, λ, β: weighting coefficients controlling the trade-off between the intervention strength, the preservation of other concepts, and the minimality of the perturbation.

In the context of LLM studied previously, such interventions have been used to remove specific pieces from the model's internal representation of the game state [Li et al., 2024, Karvonen, 2024]. The main observation is that, in the vast majority of cases, the model behaves as if the piece had effectively disappeared. The LLM continues to generate legal moves while correctly accounting for the absence of the removed piece. This provides evidence for the causal role of these internal representations in the model's behavior.

A main limitation of the probing approach is that the concepts under investigation must be predefined before any analysis can be performed. This

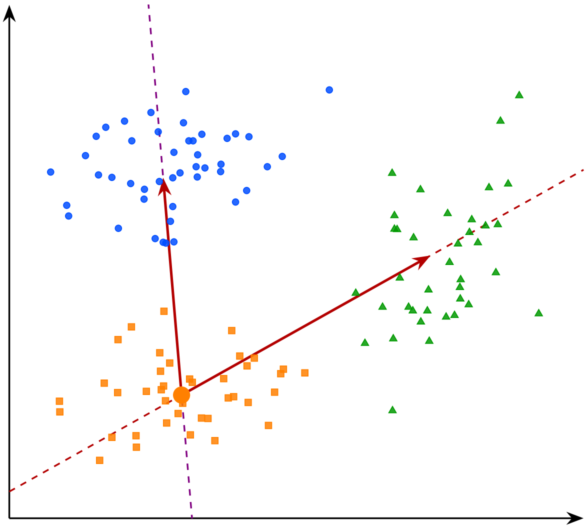


Figure 1.4: Interpretable directions in the latent space. The arrows represent semantic shifts from concept B toward concepts A and C. Moving along these directions in the latent space induces a coherent change in the model's behavior, suggesting that the transition between concepts is continuously encoded along these axes.

introduces a strong bias in what can be studied. Moreover, annotating a dataset for each concept is extremely time-consuming and requires significant human effort. Interventions in LS are also difficult to interpret, as it is not always clear what constitutes a meaningful change in the network's output under a given perturbation.

Moreover, probing methods provide a binary view of concepts, focusing on whether a given concept is present or absent in a LR. However, this binary perspective can be overly restrictive, as concepts may be represented in a more continuous or graded manner, with varying degrees of presence. To obtain a more detailed characterization of these representations, other approaches have been proposed, such as the analysis of interpretable directions in LS.

1.3 Interpretable Directions

To overcome the limitations of binary concept detection in probing methods, several approaches have been proposed to analyze the structure of LR. Among them, the notion of interpretable directions has emerged as a natural extension, aiming to capture the continuous nature of how concepts are encoded in representation spaces. Empirically, it has been observed that shifting a LR along a direction can induce coherent and semantically meaningful changes in the model's output [Haas et al., 2024, Voynov and Babenko, 2020]. Such transformations can be mathematically express as:

$$h' = h + \alpha \mathbf{v}_i,$$

where:

- h denotes the original embedding,

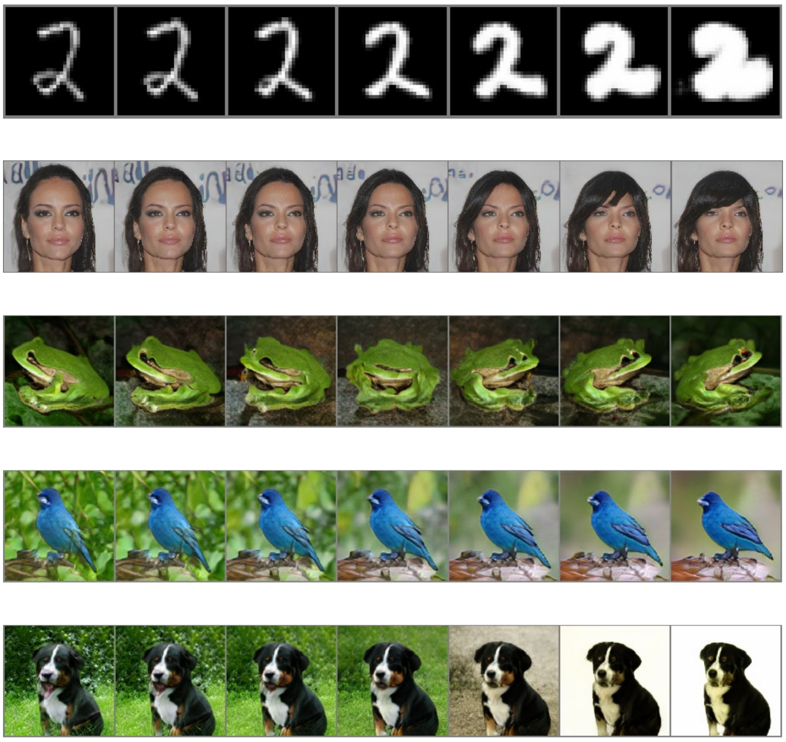


Figure 1.5: Interpretable directions in the latent space of a generative model [Voynov and Babenko, 2020]. Each row corresponds to a distinct direction $\mathbf{v}_i$, and each column to an increasing value of α. The resulting transformations are semantically coherent and diverse: stroke thickness for digits, head rotation for faces, texture and background for natural images.

- $\mathbf{v}_i$ is a unit vector associated with concept i,
- α controls the strength of the intervention,
- h' is the resulting perturbed representation.

In many cases, the resulting change in the model's output varies smoothly with α, suggesting an approximately linear structure of the LS with respect to certain semantic factors, as illustrated in Figure 1.5. This observation has led to the hypothesis that neural representations may organize concepts in a partially linear manner within the embedding space. Vectors of this form are referred to as interpretable directions, as they correspond to transformations that can be associated with specific semantic attributes. The objective of this line of work is therefore to identify directions in LS that align with meaningful variations in the encoded concepts.

This phenomenon has been extensively studied in generative image models. As illustrated in Figure 1.5, controlled perturbations of LR along interpretable directions can induce specific semantic transformations, such as stroke thickness modification in digit images, head rotation in facial images, or texture and

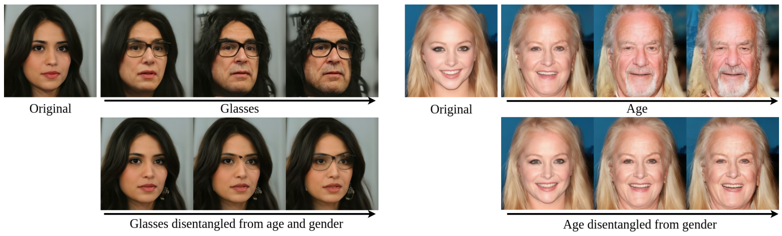


Figure 1.6: Illustration of entanglement in the latent space of a generative face model [Haas et al., 2024]. The top rows show entangled directions: moving along the glasses direction simultaneously modifies age and gender, and moving along the age direction also alters gender. This reflects the entanglement phenomenon, where a single direction in latent space induces unintended changes across multiple semantic concepts at once. The bottom rows show the corresponding disentangled directions, where each axis controls a single semantic attribute while leaving the others unchanged.

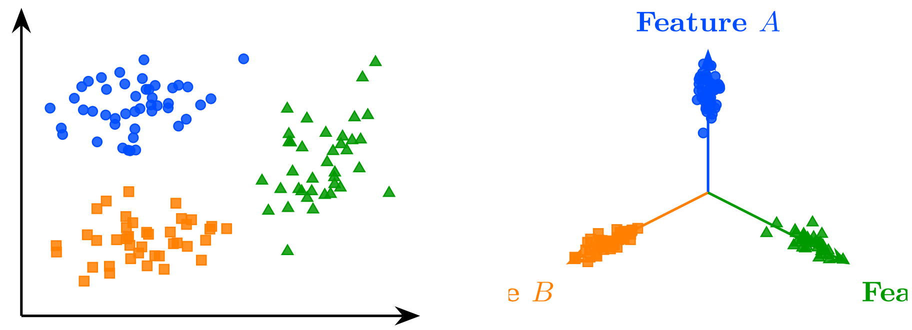
background changes in natural images [Voynov and Babenko, 2020]. These observations have led to the development of the field of LS editing, whose objective is to manipulate generated content directly through modifications in LS rather than through changes in the input prompt.

However, a central challenge remains: identifying such interpretable directions is far from trivial. Most existing approaches rely on auxiliary models that encode human-defined semantics, such as CLIP [Haas et al., 2024], to relate latent variations to textual concepts. Consequently, these methods inherit a limitation similar to that of probing techniques 1.2, since concept discovery depends on a pre-existing semantic framework defined by humans.

To overcome this limitation, a research direction known as unsupervised semantic discovery has emerged. Its goal is to uncover interpretable directions in LS without explicit semantic supervision. In these approaches, one model applies perturbations along random directions in LS, while a second model attempts to recognize the applied transformation. If a stable correspondence emerges between the perturbation and its prediction, this suggests that the considered direction corresponds to a coherent structure in LS [Voynov and Babenko, 2020]. In this framework, human intervention occurs only a posteriori, to verify whether the discovered directions indeed correspond to interpretable semantic variations.

However, despite these advances, several challenges remain. It appears that not all concepts encoded by neural networks are linearly identifiable in LS. In practice, interpretable direction methods typically discover only a few dozen directions per dataset, which intuitively seems insufficient to account for the full diversity of semantic variations that a generative model is capable of producing.

In addition, LR often exhibit a phenomenon known as entanglement, illustrated in Figure 1.6, where multiple semantic attributes are mixed within the same directions of the representation space. Ideally, one would like each semantic factor to vary independently along a specific direction, without affecting other attributes. In practice, this is rarely the case: a single direction in LS often influences multiple semantic simultaneously, meaning that modifying one concept unintentionally alters several others. This lack of separability indicates that semantic information is not cleanly factored in the representation space, which



(a) Dense latent space **(b)** Sparse SAE space

Figure 1.7: Transformation from a dense latent space to a sparse SAE-style representation. On the left **(a)**, embeddings are entangled in the initial latent space. On the right **(b)**, the sparse representation aligns each concept with a dedicated feature axis.

motivates the search for more structured decompositions.

The limited number of linearly identifiable concepts, combined with the difficulty of isolating independent semantic factors, has motivated a stronger hypothesis about how neural networks organize information internally: the superposition hypothesis. In this context, a preferred analytical tool for investigating this hypothesis is the use of Sparse AutoEncoders (SAE), which can extract concepts from neural representations.

1.4 Sparse Autoencoder

Approaches based on latent directions face inherent limitations. Empirically, only a restricted subset of concepts appears to admit a linear representation, and even these directions are often affected by significant entanglement with other concepts, illustrated in Figure 1.6.

To explain this phenomenon, a complementary hypothesis has been introduced, extending the assumptions underlying the Concept Hypothesis (1.1). This hypothesis, known as the Superposition Hypothesis [Elhage et al., 2022], posits that concepts are preferentially represented in a linear fashion, with each concept ideally corresponding to a distinct direction in LS. In this idealized setting, concept vectors would form an approximately orthogonal basis, where each direction independently controls a single factor of variation. Furthermore, only a small subset of the available concepts is expected to be active for a given representation. As a result, LR can be viewed as sparse combinations of concept directions. This idealized representation is referred to as a non-superposed representation and can be mathematically expressed as follows:

$$\mathbf{h} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i, \qquad \forall i \neq j, \quad \mathbf{v}_i^\top \mathbf{v}_j \approx 0, \qquad \|\boldsymbol{\alpha}\|_0 \ll n,$$

where:

- $\mathbf{h}$ denotes a LR,

Figure 1.8: A concept direction corresponding to the Golden Gate Bridge, discovered by a sparse autoencoder applied to a large language model [Templeton et al., 2024]. This direction activates strongly whenever the Golden Gate Bridge is evoked in the input, whether through English text, other languages, or even images, demonstrating that the learned concept is not tied to a specific surface form but captures a deeper semantic abstraction shared across modalities.

- n is the total number of concepts in the dictionary,
- $\mathbf{v}_i$ represents the direction associated with concept c_i,
- α_i corresponds to the activation strength of concept c_i in the representation $\mathbf{h}$,
- $\mathbf{v}_i^\top \mathbf{v}_j \approx 0$ indicates that concept directions are approximately orthogonal, meaning each concept independently controls a single factor of variation,
- $\|\boldsymbol{\alpha}\|_0 \ll n$ indicates that only a small fraction of concepts are active for a given representation.

According to the Superposition Hypothesis, latent representations do not perfectly follow this idealized structure because the representational capacity of NN is limited. For a given layer, the dimensionality of the representation space is finite, which constrains the number of concepts that can be encoded independently. As a result, the network cannot allocate a dedicated direction to every concept, and multiple concepts end up compressed into shared dimensions [Bricken et al., 2023]. This phenomenon is closely related to the entanglement observed in LS analyses: where entanglement is an empirical observation about the mixing of semantic attributes, the Superposition Hypothesis provides a theoretical explanation for why such mixing emerges.

Under this hypothesis, the phenomenon of superposition becomes an unavoidable obstacle for any method that attempts to analyze LR directly in the original activation space. Motivated by this view, and with the goal of recovering the idealized non-superposed, a large body of work has focused on developing methods to desuperpose LR. To this end, a common strategy consists in projecting LR into a higher-dimensional space while imposing additional constraints designed to recover a non-superposed representation. A widely used implementation of this idea is the SAE, which can be defined as follows:

$$z = f_{\text{enc}}(h), \quad \hat{h} = f_{\text{dec}}(z)$$

$$\mathcal{L}_{\text{auto}} = \gamma \|h - \hat{h}\|_2^2 + \lambda \|z\|_1$$

where:

- $f_{\text{enc}}(h)$ is the encoding function that projects the original representation h into a higher-dimensional sparse space,
- $f_{\text{dec}}(z)$ is the decoding function that reconstructs the original representation from the sparse code,
- $z \in \mathcal{Z}$ is the sparse representation of h, with $\dim(z) \gg \dim(h)$,
- $\hat{h}$ is the reconstructed representation,
- $\gamma\|h - \hat{h}\|_2^2$ is the reconstruction loss ensuring that the original representation can be faithfully recovered,
- $\lambda\|z\|_1$ is the sparsity regularization term that encourages most components of z to be zero,
- γ, λ are weighting coefficients controlling the trade-off between reconstruction fidelity and sparsity.

In practice, both f_{enc} and f_{dec} are implemented as linear projections into the high-dimensional space $\mathcal{Z}$. This choice is motivated by the same reasoning as in probing methods (Section 1.2). Using overly expressive projection functions would introduce additional bias, making it difficult to attribute the structure of the learned representations to the model itself rather than to the projection. The parameters of these projection functions are learned by minimizing $\mathcal{L}_{\text{auto}}$ using gradient-based optimization, as introduced in Section **??**. Under this formulation, each dimension of $\mathcal{Z}$ is interpreted as a dedicated concept direction. The corresponding basis vector serves as the support for a distinct concept encoded by the model.

The method proves to be powerful when successfully applied to real language models [Templeton et al., 2024]. In these applications, the idea is to desuperpose LLM LR using SAE in order to recover a non-superposed representation. Once this representation is obtained, it has been shown that many concept vectors exhibit strong semantic properties. A well-known example is the Golden Gate Bridge concept direction, illustrated in Figure 1.8, which activates systematically whenever the input text or image refers to the Golden Gate Bridge, regardless of the language or modality.

It is possible to artificially activate this concept direction by directly modifying the sparse representation z, even when the input does not contain any reference to it. This operation constitutes an explicit intervention on the model's latent representations. The edited representation is then projected back into the original activation space and the model proceeds with standard next-token generation from h^*:

$$h^* = f_{\text{dec}}(z^*), \quad \text{where} \quad z^* = z + \delta\, e_i,$$

where:

- z is the original sparse representation of the input,
- z^* is the modified sparse representation,

- e_i is the basis vector of $\mathcal{Z}$ associated with the target concept direction,
- δ controls the strength of the artificial activation,
- h^* is the resulting edited representation in the original activation space.

Empirically, it is observed that even when the prompt is unrelated to the Golden Gate Bridge, the model's output shifts to discussing it. This provides evidence for the causal role of these learned features. This type of intervention has been used by Anthropic researchers to steer or suppress specific model behaviors, effectively enabling a form of behavior control through latent-space editing.

Despite their empirical success, SAE exhibit several important limitations that question some of their underlying assumptions. These limitations can be summarized as follows:

- **Instability of the decomposition.** Repeating the SAE training process multiple times on the same model does not yield the same decomposition of the LS. Different runs produce different concept directions in $\mathcal{Z}$, even when achieving comparable reconstruction performance, which raises questions about the uniqueness and reliability of the discovered structure.
- **Lack of semantic coverage.** In the high-dimensional space $\mathcal{Z}$, a large fraction of dimensions do not correspond to any interpretable semantic concept. This abundance of semantically vacuous directions questions the extent to which the decomposition genuinely reflects the conceptual structure of the model.
- **Sparsity–semantics trade-off.** Increasing the sparsity coefficient λ beyond a certain threshold degrades the semantic quality of the learned features [Jermyn et al., 2024]. The representations tend to lose meaningful structure, suggesting that enforcing excessive sparsity comes at the cost of interpretability.
- **Failure of scaling to remove superposition.** Under the Superposition Hypothesis, increasing the dimensionality of the representation space should reduce superposition. However, this is not observed in practice. Even in very high-dimensional LS, representations remain partially superposed, indicating that scaling alone does not resolve the entanglement of features.

These limitations suggest that SAE, while powerful, do not fully resolve the problem of representational structure, and that convincing empirical results do not necessarily validate the underlying theoretical assumptions. Indeed, probing (Section 1.2), interpretable directions (Section 1.3), and SAE all rest on a shared assumption: that semantic structure can be recovered through approximately linear decompositions of LR. This assumption has driven significant empirical progress, yet it remains theoretically fragile. The following section therefore introduces a complementary approach that relaxes this constraint entirely, requiring no assumption about the linear organization of concepts: deep clustering.

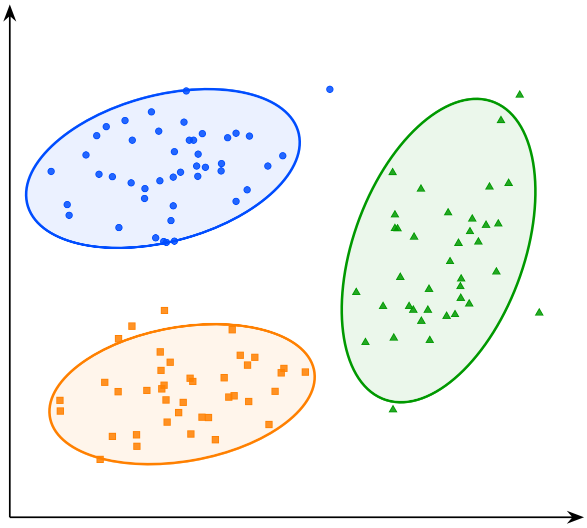


Figure 1.9: Visualization of the latent space structure. The points represent embeddings associated with concepts A, B, and C. The ellipses indicate the approximate extent of each group in the latent space and highlight their geometric separation.

1.5 Clustering

The LS analysis methods presented so far all rely, either explicitly or implicitly, on assumptions about the linear organization of concepts within the LS. For probing methods, concepts must be at least partially linearly separable for simple probes to succeed 1.2. Interpretability through latent directions assumes that variations associated with a concept can be captured by moving along a linear axis of the LS 1.3. Similarly, the superposition hypothesis assumes that concepts are represented through linear directions that become entangled because of representational capacity constraints 1.4. Although these assumptions have led to numerous empirical successes, their theoretical justification remains limited. More importantly, several observations suggest that the linearity hypothesis may not fully capture the structure of LR.

For instance, some concepts can only be recovered using non-linear probes. Likewise, methods based on interpretable directions typically identify only a limited number of semantic factors. Finally, in sparse autoencoder approaches, enforcing excessive sparsity often degrades reconstruction quality and interpretability, suggesting that the underlying representation may not be perfectly described by a sparse linear decomposition. Taken together, these observations indicate that semantic concepts may not always be organized according to a simple linear structure. This motivates the exploration of alternative approaches that rely on weaker assumptions regarding the geometry of LS.

To the best of our knowledge, clustering-based analysis is the only family of LS analysis methods that does not rely on any assumption of linear structure, which is why the following section is devoted to its presentation [Ren et al., , Wei et al.,]. Rather than assuming that concepts correspond to directions or linear subspaces, clustering methods only assume that semantically similar inputs produce nearby

Figure 1.10: Clusters discovered by a deep clustering model applied to natural images [Xie et al.,]. Each row contains the ten highest-scoring elements of a single cluster. The visual coherence within each row, grouping animals, vehicles, and aircraft into distinct clusters, suggests that the learned latent representations capture semantically meaningful structure without any explicit supervision.

LR. Conversely, inputs that differ significantly from a semantic perspective are expected to be located farther apart in the LS. This is a considerably weaker assumption, in the sense that it is empirically verified in a wide range of settings.

Under this assumption, the discovery of concepts can be reformulated as the identification of groups of embeddings that occupy the same region of the LS. Concepts are therefore no longer associated with directions, but with clusters of neighboring representations, as illustrated in Figure 1.10. This idea forms the basis of the research field known as deep clustering. In these approaches, a collection of inputs is first projected into a LS using a NN. The resulting embeddings form a point cloud on which standard clustering algorithms can be applied. The obtained clusters are then interpreted as representing distinct semantic concepts.

Deep clustering has been particularly successful in unsupervised image classification. Typically, an autoencoder is composed of an encoder f_{enc} and a decoder f_{dec}. The encoder maps an input $x^{(i)} \in \mathcal{X}$ to a low-dimensional LR $z^{(i)} \in \mathbb{R}^d$:

$$z^{(i)} = f_{\text{enc}}(x^{(i)}), \quad d \ll \dim(\mathcal{X}).$$

To ensure that this compressed representation does not discard essential information, the decoder reconstructs the input from the LS:

$$\hat{x}^{(i)} = f_{\text{dec}}(z^{(i)}) = f_{\text{dec}}(f_{\text{enc}}(x^{(i)})).$$

The model is trained to minimize the reconstruction error between the original input and its reconstruction, typically using a Mean Squared Error (MSE) loss:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N}\sum_{i=1}^{N} \left\| x^{(i)} - \hat{x}^{(i)} \right\|_2^2 .$$

This objective encourages the LS to preserve the most informative features of the input while enforcing a compact representation.

Clustering algorithms are then applied to the learned embeddings.

Let $\mathcal{D} = \{x^{(i)}\}_{i=1}^{N}$ denote a dataset of inputs. Through the encoder f_{enc}, this dataset induces a point cloud in the LS:

$$\mathcal{Z} = \{z^{(i)} \in \mathbb{R}^d \mid z^{(i)} = f_{\text{enc}}(x^{(i)}),\ x^{(i)} \in \mathcal{D}\}.$$

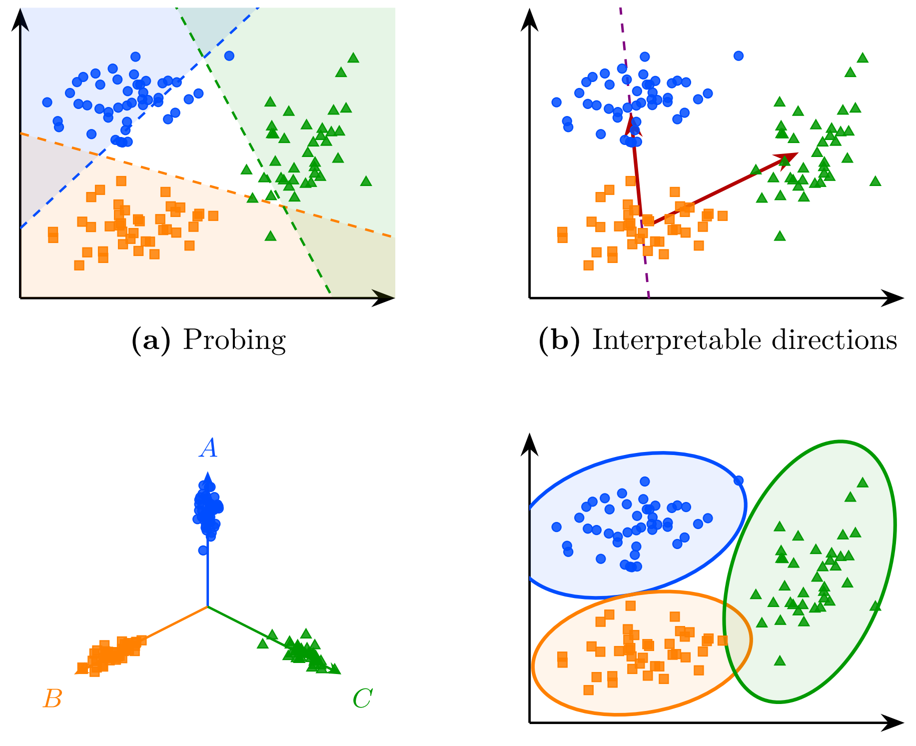
Clustering algorithms are then applied to this set $\mathcal{Z}$ of embeddings in order to identify groups of semantically similar representations.

An example of this methodology can be observed in Figure 1.10, where the approach is applied to the STL-10 dataset [Coates et al., 2011] using k-means clustering [MacQueen, 1967] with a number of clusters arbitrarily fixed to $K = 10$ [Xie et al.,]. For each cluster, the corresponding line shows the 10 samples closest to the associated centroid in the LS.

We observe that the method captures coherent semantic categories such as animals, vehicles, and flying entities. This result is particularly striking given that the only available information to the system consists of raw images. It suggests that the compression process induces a LS in which semantically meaningful structures naturally emerge, and that these structures correspond to high-level concepts.

More advanced approaches combine representation learning and clustering objectives within a single optimization process. In this setting, additional loss terms encourage embeddings to organize themselves around cluster centroids during training. This often produces LS that are easier to analyze and improves the quality of the discovered semantic structure [Guo et al., , Miklautz et al.,].

Despite its effectiveness, clustering-based analysis presents several important limitations. First, as with most unsupervised semantic discovery techniques, the interpretation of the resulting clusters remains partially subjective: while benchmark datasets often provide class labels that facilitate evaluation, there is no guarantee that the discovered clusters correspond to the categories that humans consider most relevant, and alternative semantic groupings may be equally valid. Second, clustering methods are highly sensitive to design choices, as the number of clusters, the distance metric, and the clustering algorithm itself can all significantly influence the results, making parameter selection challenging and often requiring substantial empirical tuning. Finally, to the best of our knowledge, no intervention technique has yet been proposed in the context of clustering-based analysis: unlike probing or SAE, which allow targeted modifications of LR to study their causal role, clustering methods currently offer no mechanism to alter the model's behavior by acting directly on the discovered structure.



(a) Probing

(b) Interpretable directions

(c) Sparse autoencoder

(d) Clustering

Figure 1.11: Overview of the four latent space analysis methods studied in this chapter, applied to the same point cloud.

1.6 Conclusion

The four families of methods reviewed in this chapter, namely probing, interpretable directions, sparse autoencoders, and clustering, are summarized in Figure 1.11 and all build upon the concept hypothesis introduced in Section 1.1. Each of them approaches the question from a different angle, but they share a common objective: to uncover how LR are structured and what semantic information can be extracted from them.

However, a fundamental difficulty remains. There is currently no consensus on how to rigorously evaluate to what extent these methods genuinely provide access to the internal concepts of a model. Most existing metrics assess reconstruction quality or linear separability, but these do not directly measure whether the discovered structures correspond to meaningful abstractions from the model's perspective. This leaves open the question of what it means, formally, for a concept to be represented in a NN.

Addressing this question may require contributions from disciplines beyond computer science and mathematics. As the notion of semantics is inherently tied to human interpretation, a purely formal or computational account may be insufficient. Insights from philosophy, cognitive science, or psychology could prove valuable in grounding the notion of concept in a way that is both theoretically rigorous and empirically tractable. Despite these open questions, the methods studied in this chapter provide a practical foundation for analyzing the internal representations of NNs.

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